MMDU MULLANA OFFICIAL SYLLABUS • BPHY-001 • UNIT 3

Unit 3: Wave Nature of Particles & Schrödinger Equation (Quantum Mechanics)

Subject: Applied Physics (BPHY-001) • **Exam Weightage:** 24 - 28 Marks • **Study Time:** 40 mins

de Broglie hypothesis, phase vs group velocity identity vg = v_particle, Heisenberg uncertainty principle and non-existence of nuclear electrons proof, Born wave function postulate, TDSE and TISE derivations, 1D infinite box eigenfunctions and quantized energy levels proof, zero-point energy, quantum tunneling barrier penetration, and 1D harmonic oscillator.

1. de Broglie Hypothesis, Phase & Group Velocity & Uncertainty Principle

1. de Broglie Hypothesis & Matter Waves

Louis de Broglie postulated that moving matter possesses an associated dual wave character:

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mE_k}}$$

For an electron accelerated through an electric potential difference V volts:

$$\lambda = \frac{h}{\sqrt{2meV}} = \frac{1.227}{\sqrt{V}}\ \text{nm} = \frac{12.27}{\sqrt{V}}\ \text{Å}$$

For $V = 100$ V, $\lambda = 1.227$ Å, matching crystal lattice spacings (Davisson-Germer electron diffraction proof).

2. Phase Velocity vs. Group Velocity

- **Phase Velocity (v_p):** Velocity of individual monochromatic wave crest: $v_p = \omega/k$.
- **Group Velocity (v_g):** Velocity of composite wave packet envelope carrying energy: $v_g = d\omega/dk$.
- **Interrelation:**

$$v_g = v_p - \lambda\frac{dv_p}{d\lambda} \qquad v_p \cdot v_g = c^2 \quad \text{(Relativistic)}$$

- **Fundamental Identity:** The group velocity of a de Broglie matter wave packet is identically equal to classical particle velocity:

> $$v_g = v_{\text{particle}}$$

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3. Heisenberg Uncertainty Principle & Nuclear Electron Proof

> $$\Delta x \cdot \Delta p_x \geq \frac{\hbar}{2} \qquad \Delta E \cdot \Delta t \geq \frac{\hbar}{2}$$

✅ **FUNDAMENTAL LAW / RULE:**
Proof: Non-Existence of Free Electrons Inside the Atomic Nucleus

> Typical nuclear diameter is $2R \approx 10^{-14}$ m. If an electron resides inside the nucleus, its position uncertainty is at most $\Delta x \approx 10^{-14}$ m.

> By Heisenberg's principle:

> $$\Delta p \geq \frac{\hbar}{2\Delta x} = \frac{1.054 \times 10^{-34}}{2 \times 10^{-14}} \approx 5.27 \times 10^{-21} \text{ kg} \cdot \text{m/s}$$

> Relativistic energy $E \approx pc$:

> $$E \approx (5.27 \times 10^{-21})(3 \times 10^{8}) \approx 1.58 \times 10^{-12} \text{ J} \approx 9.87 \text{ to } 20 \text{ MeV}$$

> Experimental β-decay reveals emitted electrons possess energies of only 2 to 3 MeV. Hence, **electrons cannot exist as**

> **static constituents of the nucleus**.

2. The Wave Function & Born Interpretation

1. Physical Meaning of Wave Function ψ

The wave function $\psi(\vec{r}, t)$ is a complex probability amplitude. Max Born formulated its statistical interpretation:

$$P(\vec{r}, t)dV = |\psi(\vec{r}, t)|^2 dV = \psi^* \psi \, dV$$

Well-Behaved Wave Function Conditions:

1. ψ must be **finite and continuous** everywhere.
2. ψ must be **single-valued** at all spatial coordinates.
3. The spatial gradient $\frac{\partial \psi}{\partial x}$ must be continuous across boundaries.
4. ψ must be **square-integrable (normalizable)**:

$$\int_{-\infty}^{+\infty} |\psi|^2 dV = 1$$

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2. Quantum Operators & Expectation Values

Observable	Quantum Operator	Expectation Value Formula
Position x	$\hat{x} = x$	$\langle x \rangle = \int \psi^* x \psi \, dx$
Momentum p_x	$\hat{p_x} = -i\hbar \frac{\partial}{\partial x}$	$\langle p_x \rangle = \int \psi^* \left(-i\hbar \frac{\partial}{\partial x}\right) \psi \, dx$
Energy E	$\hat{E} = i\hbar \frac{\partial}{\partial t}$	$\langle E \rangle = \int \psi^* \left(i\hbar \frac{\partial}{\partial t}\right) \psi \, dx$

| **Hamiltonian** H | $\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + V$ | $\langle H \rangle = \int \psi^* \hat{H} \psi \, dV$

3. Derivation of Schrödinger Equations (TDSE & TISE)

1. 1D Time-Dependent Schrödinger Equation (TDSE)

A free particle propagating along $+x$ is represented by plane wave:

$$\psi(x,t) = Ae^{i(kx-\omega t)} = Ae^{i(px-Et)/\hbar}$$

- Differentiating with respect to t: $\frac{\partial \psi}{\partial t} = -\frac{iE}{\hbar}\psi \implies E\psi = i\hbar\frac{\partial \psi}{\partial t}$

- Differentiating twice with respect to x: $\frac{\partial^2 \psi}{\partial x^2} = -\frac{p^2}{\hbar^2}\psi \implies p^2\psi = -\hbar^2\frac{\partial^2 \psi}{\partial x^2}$

Total classical energy $E = \frac{p^2}{2m} + V(x,t)$. Operating on ψ:

$$i\hbar\frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2 \psi}{\partial x^2} + V(x,t)\psi \quad \Leftrightarrow \quad i\hbar\frac{\partial \psi}{\partial t} = \hat{H}\psi$$

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2. 1D Time-Independent Schrödinger Equation (TISE)

Using separation of variables $\psi(x,t) = \phi(x)e^{-iEt/\hbar}$:

$$i\hbar\left(-\frac{iE}{\hbar}\right)\phi e^{-iEt/\hbar} = \left[-\frac{\hbar^2}{2m}\frac{d^2\phi}{dx^2} + V(x)\phi\right]e^{-iEt/\hbar}$$

$$\frac{d^2\phi}{dx^2} + \frac{2m}{\hbar^2}[E - V(x)]\phi = 0$$

4. Particle in a 1D Infinite Well & Quantum Barrier Tunneling

1. Particle in a 1D Infinite Box (Rigid Well Derivation)

Potential: $V(x) = 0$ for $0 \le x \le L$; $V(x) = \infty$ for $x < 0, x > L$.
Inside well: $\frac{d^2\psi}{dx^2} + k^2\psi = 0$ where $k^2 = \frac{2mE}{\hbar^2}$.
General solution: $\psi(x) = A\sin(kx) + B\cos(kx)$.

1. Boundary condition $\psi(0) = 0 \implies B = 0$.

2. Boundary condition $\psi(L) = 0 \implies A\sin(kL) = 0 \implies k_nL = n\pi$ $(n = 1, 2, 3...)$.

$$k_n = \frac{n\pi}{L} \implies E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{n^2\pi^2\hbar^2}{2mL^2} = \frac{n^2h^2}{8mL^2}$$

Normalizing $\int_0^L |\psi_n|^2 dx = 1 \implies A = \sqrt{\frac{2}{L}}$:

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$$

✅ **FUNDAMENTAL LAW / RULE:**
Zero-Point Energy ($n = 1$):

$$E_1 = \frac{h^2}{8mL^2} \neq 0$$

A particle in a box cannot have zero energy, because zero energy implies exact zero momentum ($\Delta p = 0$), violating the Heisenberg uncertainty principle!

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2. Quantum Tunneling & 1D Harmonic Oscillator

- **Barrier Penetration:** Particle of energy $E < V_0$ penetrating a rectangular barrier exhibits decaying wavefunction $\psi(x) \sim e^{-\alpha x}$.

$$T \approx e^{-2\alpha a} \quad \text{where } \alpha = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$$

- **1D Quantum Harmonic Oscillator:** Parabolic potential $V(x) = \frac{1}{2}m\omega^2 x^2$:

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega \quad (n = 0, 1, 2...)$$

Zero-point energy $E_0 = \frac{1}{2}\hbar\omega$; energy levels are strictly equidistant with spacing $\Delta E = \hbar\omega = hv$.